

Stochastic Control in Finance: From HJB Validation to Dynamic Hedging under Transaction Costs and Barrier Risk

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Abstract. This paper studies stochastic control in finance through a sequence of theoretical, numerical, and empirical experiments. The first contribution validates a finite-difference Hamilton–Jacobi–Bellman (HJB) solver on the one-dimensional Merton portfolio problem, for which a closed-form value function and optimal feedback are available. The numerical policy matches the analytic solution with interior control RMSE as low as 1.1×10^{-5} for moderate risk aversion. The second experiment tests the Merton allocation rule on AAPL using walk-forward estimates of drift and volatility. The exercise highlights a central practical limitation: an analytically optimal rule can become unstable when the drift must be estimated, particularly when leverage is unconstrained. The third experiment turns to option hedging. A discrete Bellman formulation recovers the Black–Scholes delta in the frictionless European-call case without using the delta during optimization; proportional transaction costs then generate endogenous BUY/HOLD/SELL regions. Finally, the framework is extended to a down-and-out call. A Brownian-bridge treatment of barrier crossings and a survival-state variable allow the Bellman policy to adapt strongly near the knock-out boundary. The results illustrate a progression from closed-form stochastic control to numerically challenging hedging problems, while clearly separating validated results from exploratory multidimensional branches.

Keywords: stochastic control; Hamilton–Jacobi–Bellman equation; Merton portfolio problem; dynamic programming; option hedging; transaction costs; barrier options; Brownian bridge.

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1. Introduction

Many financial decisions are dynamic: a portfolio is rebalanced as prices evolve, and an option hedge is adjusted as the underlying moves. Stochastic control provides a mathematical language for such problems. A state process evolves randomly, a controller chooses an adapted action, and an objective functional ranks the resulting trajectories. Dynamic programming then converts the global optimization problem into a local recursion, or, in continuous time, an HJB equation [5, 6].

This work is organized around four experiments of in-

creasing financial complexity. The first asks whether a numerical HJB method can reproduce a known analytic optimum. The second asks what happens when the same optimum is fed estimated rather than known market parameters. The third moves from investment to hedging and asks whether Bellman optimization can recover the Black–Scholes hedge and then adapt to transaction costs. The fourth introduces a path-dependent barrier product and asks whether the optimal hedge changes as the contract approaches knock-out.

The paper deliberately focuses on one coherent route—dynamic programming, HJB, and discrete Bell-

man control—rather than combining multiple unrelated stochastic-control formalisms. The main methodological principle is validation by nested benchmarks: whenever a closed-form or high-quality reference exists, the numerical controller is tested against it before more complex features are introduced. A recurring distinction throughout the study is the difference between the physical measure, under which realized asset dynamics are simulated with drift μ , and the risk-neutral measure, under which derivative-pricing benchmarks replace the drift by the short rate r . This distinction is introduced here because both measures appear repeatedly in the hedging experiments and is discussed explicitly in Section 9.3.

2. Financial Products and Market Model

2.1 Risk-free asset and risky stock

The basic market contains a money-market account B_t and a risky asset S_t . Under constant parameters,

$$dB_t = rB_t dt, \quad \frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad (1)$$

where r is the short rate, μ the physical drift, $\sigma > 0$ the volatility, and W a Brownian motion. The geometric Brownian motion model is the classical basis of Black–Scholes–Merton option theory [3, 4].

If π_t denotes the *monetary amount* invested in the risky asset and the remaining wealth $X_t - \pi_t$ is invested at the risk-free rate, self-financing gives

$$dX_t = [rX_t + \pi_t(\mu - r)] dt + \pi_t \sigma dW_t. \quad (2)$$

This equation follows directly by adding the gains on the two holdings.

2.2 European call

A European call with strike K and maturity T pays

$$H_{\text{call}} = (S_T - K)^+. \quad (3)$$

Under the frictionless Black–Scholes assumptions, the arbitrage-free price $C(t, S)$ solves the Black–Scholes PDE and its delta

$$\Delta^{BS}(t, S) = \frac{\partial C}{\partial S}(t, S) = N(d_1) \quad (4)$$

is the classical continuous-time replicating position [3, 4].

2.3 Down-and-out call

A down-and-out call is a barrier option. It behaves like a call only if the underlying has remained above a lower barrier B throughout the contract life:

$$H_{\text{DOC}} = (S_T - K)^+ \mathbf{1}_{\{\min_{0 \leq u \leq T} S_u > B\}}. \quad (5)$$

The payoff is therefore path dependent. Once the barrier is hit, the liability is permanently extinguished. Barrier monitoring is numerically delicate: checking only discrete endpoints misses crossings between observation times. Brownian-bridge conditioning is a standard remedy in Monte Carlo barrier simulation [8, 9].

2.4 Transaction costs

If the current stock position is q and it is changed to a , the experiments impose proportional cost

$$C_{\text{tr}} = \lambda S |a - q|, \quad (6)$$

with cost rate $\lambda \geq 0$. Transaction costs break exact continuous replication and create a trade-off between tracking error and turnover [10, 11, 12].

3. Stochastic Control and Dynamic Programming

3.1 Value function

For a controlled diffusion

$$dX_s = b(s, X_s, \alpha_s) ds + \sigma(s, X_s, \alpha_s) dW_s, \quad (7)$$

and objective

$$J(t, x; \alpha) = \mathbb{E} \left[\int_t^T f(s, X_s, \alpha_s) ds + g(X_T) \right], \quad (8)$$

the value function is

$$V(t, x) = \sup_{\alpha \in \mathcal{A}} J(t, x; \alpha). \quad (9)$$

It is not a terminal utility itself: it is the best future objective attainable from state (t, x) . At maturity, no decision remains, so $V(T, x) = g(x)$.

The dynamic programming principle states that an optimal policy remains optimal after any intermediate date. Applying Itô's formula and optimizing the local drift yields the HJB equation [5, 6]

$$V_t + \sup_{\alpha \in \mathcal{A}} \left\{ b(t, x, \alpha) V_x + \frac{1}{2} \sigma^2(t, x, \alpha) V_{xx} + f(t, x, \alpha) \right\} = 0. \quad (10)$$

3.2 CRRA utility and the Merton problem

The Merton allocation experiment uses constant-relative-risk-aversion utility

$$U(x) = \frac{x^{1-\gamma}}{1-\gamma}, \quad \gamma > 0, \gamma \neq 1, \quad (11)$$

where γ controls risk aversion. The objective is

$$V(t, x) = \sup_{\pi} \mathbb{E} [U(X_T) | X_t = x]. \quad (12)$$

For wealth dynamics (2), the HJB equation is

$$V_t + \sup_{\pi} \left\{ (rx + \pi(\mu - r)) V_x + \frac{1}{2} \pi^2 \sigma^2 V_{xx} \right\} = 0. \quad (13)$$

The first-order condition gives

$$\pi^*(t, x) = -\frac{\mu - r}{\sigma^2} \frac{V_x}{V_{xx}}. \quad (14)$$

For CRRA utility, the exact solution is [1, 2]

$$V(t, x) = \frac{x^{1-\gamma}}{1-\gamma} \exp \left[(1-\gamma) \left(r + \frac{(\mu - r)^2}{2\gamma\sigma^2} \right) (T - t) \right], \quad (15)$$

$$\pi^*(t, x) = \frac{\mu - r}{\gamma\sigma^2} x. \quad (16)$$

Thus the optimal risky *fraction* is constant: $w^* = \pi^*/x = (\mu - r)/(\gamma\sigma^2)$.

3.3 Discrete Bellman hedging

For hedging, the experiments use a discrete-time Bellman recursion. The control a_n is the new number of shares held after rebalancing. With current cash b_n , stock position q_n , and stock price S_n ,

$$b_n^+ = b_n - (a_n - q_n)S_n - \lambda S_n |a_n - q_n|, \quad (17)$$

$$b_{n+1} = b_n^+ e^{r\Delta t}, \quad q_{n+1} = a_n. \quad (18)$$

For a terminal liability H , the loss is $(W_T - H)^2$ with $W_T = b_T + q_T S_T$. Hence

$$V_n(z) = \min_a \mathbb{E} [V_{n+1}(Z_{n+1}) \mid Z_n = z, a_n = a]. \quad (19)$$

The corresponding optimizer $a_n^*(z)$ is the feedback hedge. Importantly, the Bellman solvers in Experiments 3B and 4 do not call an analytic delta during optimization; deltas are used only afterward for validation.

When $\lambda = 0$ and terminal loss is quadratic, the European-call Bellman problem admits the exact-in-wealth representation

$$V_n(S, W) = A_n(S)W^2 - 2B_n(S)W + C_n(S), \quad (20)$$

which eliminates the cash grid and permits continuous quadratic minimization in the control. This structural reduction is central to the high-accuracy frictionless validation.

4. Numerical Methods

4.1 HJB finite differences

Experiment 1 works in log-wealth $y = \log x$ on a uniform grid and marches backward from $V(T, x) = U(x)$. Spatial derivatives are approximated by centered differences in the interior. The numerical control is reconstructed from (14). Concavity diagnostics monitor $V_{xx} < 0$, and interpolation is used to evaluate $V(0, x_0)$ exactly at $x_0 = 1$ rather than at the nearest grid point. Finite-difference methods are standard for low-dimensional controlled diffusions, although the curse of dimensionality rapidly limits grid methods [7, 6].

4.2 Walk-forward parameter estimation

Experiment 2 uses adjusted AAPL prices from 2016–2025 and a three-month U.S. Treasury rate proxy. At each monthly decision date, only the previous 504 daily log returns are used. With R denoting daily log returns,

$$\hat{\sigma}_t = \text{sd}(R)\sqrt{252}, \quad \hat{\mu}_t = 252 \bar{R} + \frac{1}{2}\hat{\sigma}_t^2, \quad (21)$$

and

$$\hat{w}_t^* = \frac{\hat{\mu}_t - r_t}{\gamma \hat{\sigma}_t^2}. \quad (22)$$

Both unconstrained and long-only $[0, 1]$ versions are tested. This experiment is not another HJB validation; it is a stress test of the Merton feedback when the unknown parameters in the closed-form rule are replaced by historical estimates.

4.3 Quadrature and barrier crossing

Bellman expectations are computed with Gauss–Hermite quadrature. In the barrier experiment, if both step endpoints exceed B , the conditional survival probability is

$$p_{\text{surv}} = 1 - \exp\left[-\frac{2\log(S_n/B)\log(S_{n+1}/B)}{\sigma^2\Delta t}\right], \quad (23)$$

while survival is zero if either endpoint is at or below the barrier. This Brownian-bridge correction captures between-step barrier hits that endpoint monitoring misses [8].

5. Experiment 1: HJB Validation on Merton

5.1 Objective and setup

The first experiment is a numerical verification test. Parameters are

$$T = 1, \quad x_0 = 1, \quad r = 0.02, \quad \mu = 0.08, \quad \sigma = 0.20,$$

with $\gamma \in \{1.5, 3, 5\}$. The numerical HJB solution is compared with (15)–(16). Spatial sweeps use $N_x \in \{50, 100, 200, 400\}$ with $N_t = 1600$; temporal sweeps use $N_t \in \{50, 100, 200, 400\}$ with $N_x = 200$; the profile run uses $(N_x, N_t) = (401, 1600)$.

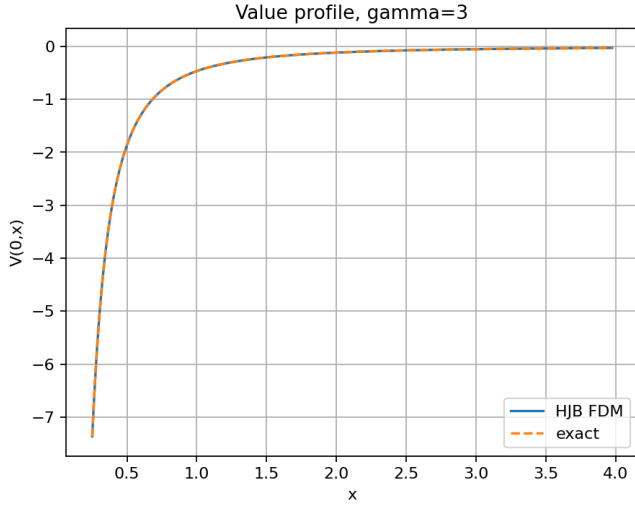
5.2 Results

Table 1: Experiment 1: profile errors at $x_0 = 1$.

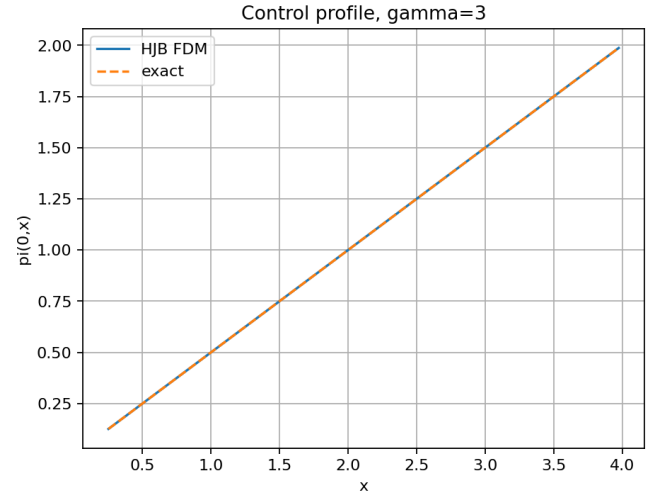
γ	rel. err. V	π_{num}	π_{exact}	RMSE π
1.5	4.53×10^{-7}	0.999997	1.000000	1.12×10^{-5}
3.0	1.35×10^{-5}	0.499950	0.500000	1.79×10^{-4}
5.0	8.25×10^{-5}	0.299856	0.300000	5.15×10^{-4}

No final-profile point violates the expected concavity and no denominator clipping is required. Table 1 and Figure 1 are generated from the same final profile run, $(N_x, N_t) = (401, 1600)$, so the tabulated errors and the visual overlay are directly consistent. For $\gamma = 3$, the exact risky allocation is $\pi^*(x) = 0.5x$.

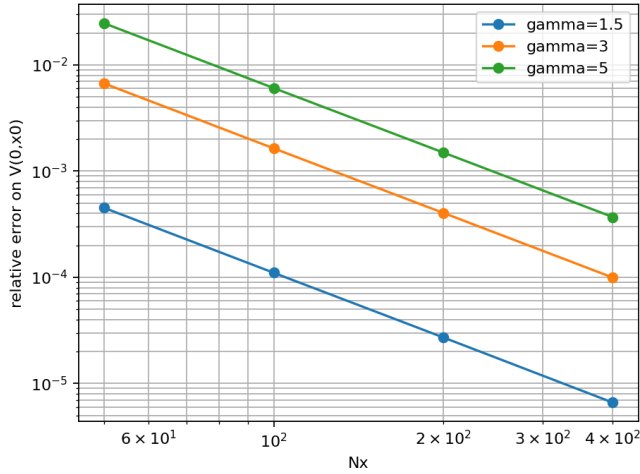
Spatial convergence is clean and monotone. For $\gamma = 3$, the relative value error decreases from 6.70×10^{-3} at $N_x = 50$ to 9.95×10^{-5} at $N_x = 400$. Temporal error is stable but not perfectly monotone because spatial truncation and explicit-scheme errors partially compensate. The experiment therefore validates the HJB implementation while also illustrating why convergence studies should isolate error sources rather than report a single fine grid.



(a) Numerical and exact value functions.



(b) Numerical and exact feedback controls.

Figure 1: Experiment 1, $\gamma = 3$: finite-difference HJB versus the Merton closed form.**Figure 2:** Experiment 1: relative value error under spatial refinement.

6. Experiment 2: Empirical Merton Allocation on AAPL

6.1 Objective

The second experiment asks a different question: does the theoretically optimal allocation remain attractive when μ and σ are not known and must be estimated from historical data? Adjusted AAPL prices and a three-month Treasury proxy are used in a no-look-ahead monthly walk-forward backtest. The test period is 1 February 2018 to 31 December 2025, with 95 monthly decisions and a 504-observation estimation window. The locally archived market series were sourced from Yahoo Finance and the Federal Reserve H.15 data program [14, 13].

6.2 Results and interpretation

AAPL itself is the strongest ex-post performer over this sample, ending at 6.93 from initial wealth one. The long-only Merton rules remain economically sensible—higher γ lowers exposure and volatility—but do not dominate

the simple benchmarks. The unconstrained rule is fragile: for $\gamma = 1.5$, the estimated weight exceeds one on 78.95% of monthly decisions and the average risky exposure is 2.30. The resulting leverage produces 182.46% annualized volatility and wealth close to zero.

This is not a contradiction of Merton optimality. Equations (15)–(16) assume known model parameters. Experiment 2 replaces the unknown drift with a noisy rolling estimate. Since w^* depends linearly on $\hat{\mu} - r$ and inversely on $\hat{\sigma}^2$, positive drift-estimation errors combined with low estimated volatility can produce extreme leverage. The result is therefore a practical lesson: parameter uncertainty, especially drift uncertainty, can dominate the elegance of the closed-form control.

The certainty-equivalent ranking reinforces the same point. AAPL has the best annualized certainty equivalent for $\gamma = 1.5$ and 3 (24.47% and 17.19%, respectively), whereas the 50/50 strategy is best for $\gamma = 5$ (10.84%). CRRA interpretation becomes problematic on paths with nonpositive wealth; accordingly, the heavily levered unconstrained case is treated with caution rather than used to claim utility superiority.

7. Experiment 3: European-Call Hedging

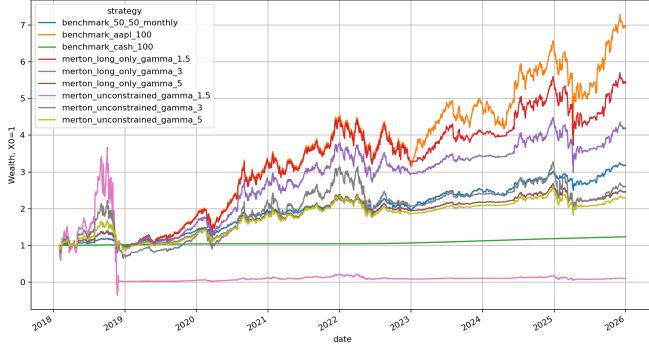
7.1 3A. Frequency and heuristic no-trade bands

The first hedging sub-experiment establishes the friction benchmark. A European call with $S_0 = K = 100$, $T = 1$, $r = 2\%$, $\sigma = 20\%$, and physical drift $\mu = 8\%$ is hedged on 20,000 exact-GBM paths. With $\lambda = 0$, more frequent discrete delta hedging reduces terminal RMSE from 1.945 at 12 rebalances to 0.436 at 252 rebalances.

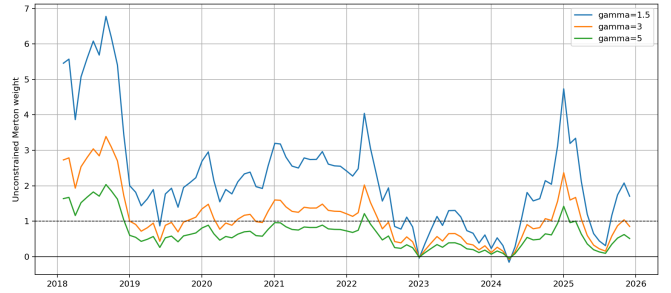
With costs, this monotonicity disappears. At $\lambda = 1\%$, the best systematic frequency is only 12 rebalances, while a no-trade rule around the Black–Scholes delta performs better with a band of 0.10. This heuristic experiment establishes the economic mechanism: trading more often improves local tracking but accumulates costs.

Table 2: Experiment 2: selected empirical performance metrics, 2018–2025.

Strategy	Final wealth	Ann. return	Ann. vol.	Sharpe	Max drawdown
Merton unconstr., $\gamma = 1.5$	0.103	-25.04%	182.46%	0.243	-109.68%
Merton long-only, $\gamma = 1.5$	5.420	23.87%	29.65%	0.780	-38.52%
Merton long-only, $\gamma = 3$	4.176	19.84%	27.59%	0.698	-38.52%
Merton long-only, $\gamma = 5$	2.453	12.04%	20.80%	0.523	-36.51%
AAPL 100%	6.933	27.79%	30.91%	0.862	-38.52%
50/50 monthly	3.175	15.76%	15.30%	0.859	-20.69%
Cash 100%	1.234	2.70%	0.13%	≈ 0	0.00%



(a) Out-of-sample wealth trajectories.



(b) Unconstrained Merton weights.

Figure 3: Experiment 2: empirical allocation illustrates the effect of parameter estimation and leverage.

7.2 3B. Bellman control

The second sub-experiment asks whether dynamic programming can discover the hedge rather than being given a delta target. In the frictionless case, the quadratic reduction (20) removes the wealth grid. The best validation run uses 48 decision times, 401 stock grid points, continuous control, and 15-point Gauss–Hermite quadrature.

The Bellman policy agrees extremely closely with the Black–Scholes delta: interior policy RMSE is 0.00311, MAE is 0.00201, and maximum interior absolute error is 0.0272. On identical 48-step simulations, hedging RMSE is 0.9652 for Bellman and 0.9874 for the discrete Black–Scholes hedge, a ratio of 0.978. The small difference is consistent with the fact that Bellman minimizes discrete physical-measure mean-square error, whereas Black–Scholes delta is the continuous-time replication benchmark.

When $\lambda > 0$, the full state (S, b, q) is retained. The Bellman optimizer produces BUY, HOLD, and SELL regions without being given a no-trade band. The mean HOLD width increases from 0.00218 at $\lambda = 0.0005$ to 0.03646 at $\lambda = 0.01$, while the realized HOLD fraction reaches 80% at the highest cost. This is the qualitative signature expected from proportional costs: small deviations from the frictionless target are not worth trading.

Quantitatively, however, the multidimensional cost branch remains less accurate than the heuristic benchmark. At $\lambda = 0.002$, Bellman terminal RMSE is 3.33, versus 1.99 for Black–Scholes at the same frequency and 1.03 for the best heuristic no-trade rule. This gap should not be interpreted as evidence that Bellman control is intrinsically inferior. At zero cost, the quadratic-in-wealth reduction eliminates the cash dimension and permits a continuous control, whereas proportional costs force the

implemented solver back to an explicit (S, b, q) grid. The additional cash and position dimensions require coarser discretizations, introduce interpolation error, and quantize the admissible control. The numerical sweeps confirm that these effects dominate before a clean multidimensional convergence regime is reached. Accordingly, at the computational resolution considered here, the cost branch is interpreted as *validated qualitatively, not fully converged quantitatively*.

8. Experiment 4: Barrier-Option Hedging

8.1 Objective, product definition, and Markov state

The final experiment extends the previous hedging framework from a standard European call to a path-dependent derivative: a *down-and-out call*. The purpose is to determine how an optimal feedback hedge changes when the liability itself may disappear before maturity.

A down-and-out call pays the usual European-call payoff only if the underlying asset has never crossed a lower barrier B . Its terminal payoff is

$$H_T = (S_T - K)^+ \mathbf{1}_{\{\min_{0 \leq u \leq T} S_u > B\}}, \quad (24)$$

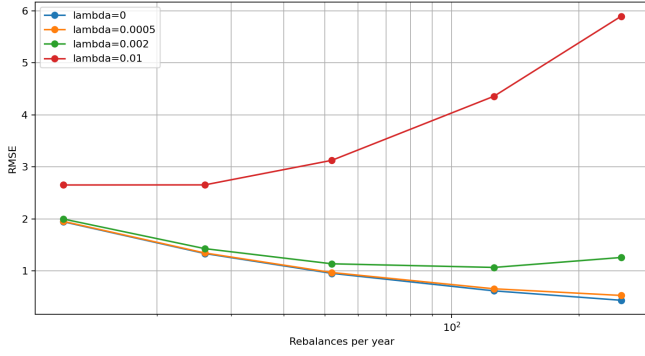
where K is the strike and B is the knock-out barrier. The experiment uses

$$B = 80, \quad S_0 = K = 100, \quad T = 1,$$

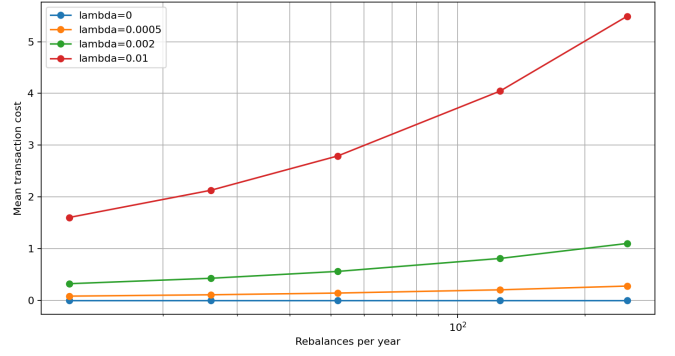
with

$$r = 2\%, \quad \mu = 8\%, \quad \sigma = 20\%.$$

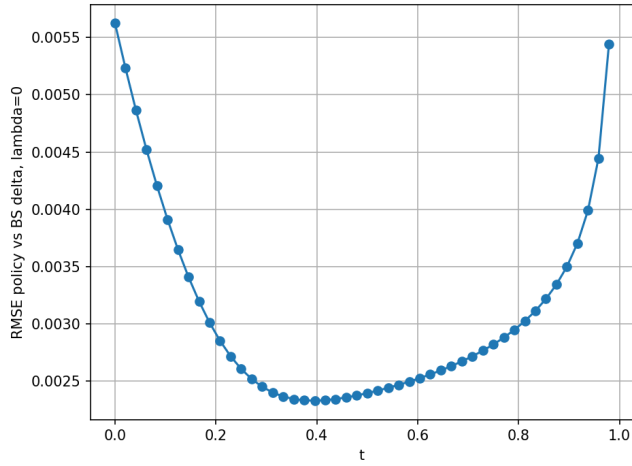
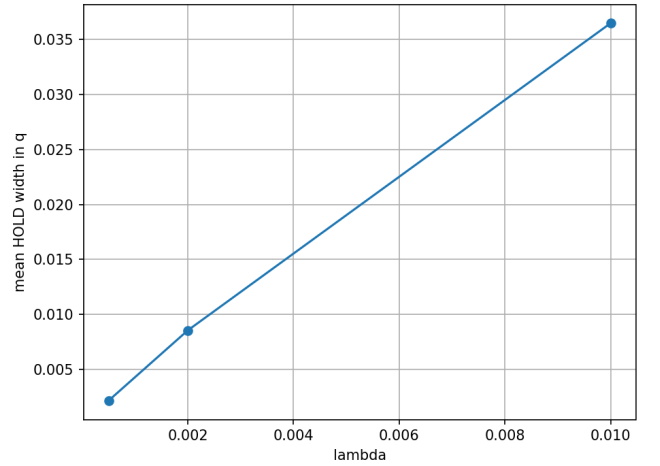
The stock follows the geometric Brownian motion in (1). Unlike a European call, however, the payoff is not determined by S_T alone: two paths ending at the same



(a) Hedging RMSE versus rebalancing frequency.



(b) Transaction cost versus frequency.

Figure 4: Experiment 3A: the frictionless benefit of frequent hedging competes with transaction costs.**Figure 5:** Experiment 3B: Bellman policy versus Black-Scholes delta at zero transaction cost.**Figure 6:** Experiment 3B: endogenous HOLD-region width increases with transaction costs.

terminal stock price may produce different liabilities if one path crossed the barrier and the other did not.

To make this path dependence compatible with dynamic programming, the state is augmented by a survival indicator

$$I_t = \begin{cases} 1, & \text{if the barrier has not been hit by time } t, \\ 0, & \text{if the option has already been knocked out.} \end{cases} \quad (25)$$

Once $I_t = 0$, the contract remains inactive permanently. The terminal payoff can therefore be rewritten in Markov form as

$$H_T = I_T(S_T - K)^+. \quad (26)$$

The pair (S_t, I_t) thus contains the past information relevant for the future liability.

The hedger controls the risky position in a self-financing portfolio. In the frictionless case, let W_t denote total wealth and let a_t be the amount of risky exposure chosen over the next decision interval. The objective is to minimize the expected squared replication error,

$$\min_{\{a_n\}} \mathbb{E} \left[(W_T - I_T(S_T - K)^+)^2 \right]. \quad (27)$$

Accordingly, the Bellman value function is

$$V_n(S, W, I) = \inf_{\{a_k\}_{k=n}^{N-1}} \mathbb{E}_n \left[(W_T - I_T(S_T - K)^+)^2 \right]. \quad (28)$$

At maturity no decisions remain, so

$$V_N(S, W, I) = [W - I(S - K)^+]^2. \quad (29)$$

Expanding the square gives

$$V_N(S, W, I) = W^2 - 2I(S - K)^+W + I((S - K)^+)^2, \quad (30)$$

because $I^2 = I$ for $I \in \{0, 1\}$. This terminal condition is quadratic in wealth. In the absence of transaction costs, the Bellman recursion preserves this structure, so at every date

$$V_n(S, W, I) = A_n(S, I)W^2 - 2B_n(S, I)W + C_n(S, I) \quad (31)$$

for coefficient functions A_n , B_n , and C_n . Numerically, this is crucial: wealth does not need to be discretized on a separate grid. The algorithm propagates the three coefficient functions backward in time and obtains the optimal hedge by minimizing the resulting quadratic Bellman objective. The barrier delta is never supplied to this optimizer; it is used only afterward as an external benchmark.

A second difficulty is barrier monitoring. The numerical algorithm observes the stock at discrete dates t_n , but a continuous path can cross B between t_n and t_{n+1} and return above the barrier before either endpoint is recorded. Endpoint-only monitoring therefore systematically misses some knock-outs. Conditional on two consecutive values $S_n > B$ and $S_{n+1} > B$, Brownian-bridge conditioning gives the crossing probability

$$p_{\text{hit}} = \exp \left[-\frac{2 \log(S_n/B) \log(S_{n+1}/B)}{\sigma^2 \Delta t} \right], \quad (32)$$

and hence the survival probability

$$p_{\text{surv}} = 1 - \exp \left[-\frac{2 \log(S_n/B) \log(S_{n+1}/B)}{\sigma^2 \Delta t} \right]. \quad (33)$$

If either endpoint is already at or below the barrier, survival is zero. This correction, consistent with the standard Brownian-bridge treatment summarized in (23), is used both inside Bellman expectations and in the Monte Carlo benchmark. The experiment therefore combines three ingredients: the survival indicator makes the path-dependent claim Markovian, the quadratic loss yields an exact-in-wealth frictionless Bellman representation, and Brownian-bridge conditioning accounts for barrier events occurring between discrete decision times.

8.2 Barrier crossing and frictionless hedge

Endpoint-only monitoring estimates a 17.30% knock-out fraction, whereas Brownian-bridge monitoring gives 18.55%, a 1.25 percentage-point increase. The difference is material and confirms that coarse endpoint monitoring understates barrier events.

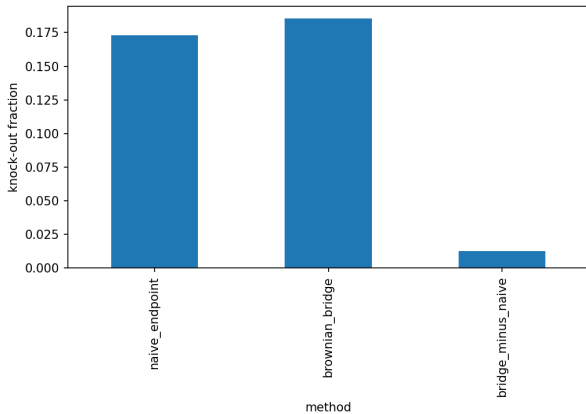


Figure 7: Experiment 4: Brownian-bridge monitoring detects additional barrier crossings.

At 24 rebalances and zero transaction costs, Bellman achieves RMSE 1.352, compared with 1.395 for the numerical barrier-delta hedge and 1.397 for the plain European delta. Temporal convergence is clear: Bellman RMSE falls from 2.479 at six decisions to 1.306 at 24 decisions. Spatial refinement beyond roughly 151 stock nodes changes little, identifying time discretization as the dominant error source in this regime.

The most informative result concerns the policy near the barrier. At $S = 100$, the Bellman position is 0.598,

nearly identical to the barrier delta 0.594. At $S = 80.8$, however, the estimated probability of knock-out before the next decision is 88.8%; Bellman cuts the hedge to 0.130, while the numerical barrier delta remains 0.342. The controller therefore recognizes that a liability with a high probability of disappearing soon should not be hedged as aggressively as a surviving vanilla call.

Table 3: Experiment 4: policy profiles at $t = 0$.

S	barrier delta	vanilla delta	Bellman q	next-step KO prob.
80.8	0.342	0.193	0.130	0.888
84.0	0.320	0.251	0.342	0.057
100.0	0.594	0.579	0.598	0.000
120.0	0.864	0.867	0.871	0.000
150.0	0.980	0.987	0.980	0.000

The transaction-cost barrier branch is exploratory. Once proportional costs are introduced, the exact quadratic reduction is no longer available in the implemented formulation and the state expands to (S, b, q, I) . To keep computation feasible, this pilot branch uses a substantially coarser grid than the frictionless experiment. Its RMSE therefore rises to approximately 5.90–8.00, compared with about 1.3–2.5 in the refined zero-cost convergence runs. The increase is consistent with the much harder four-dimensional interpolation/control problem and should not be read as a failure of the frictionless Bellman mechanism. The branch does lower turnover, but it does not yet display a clean monotone HOLD response to λ ; consequently, no claim of converged optimality is made for this branch.

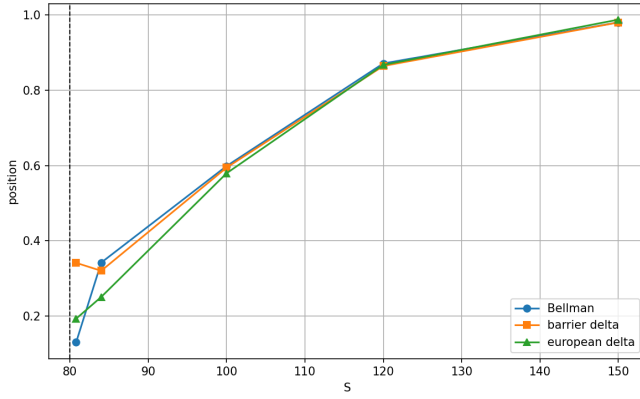
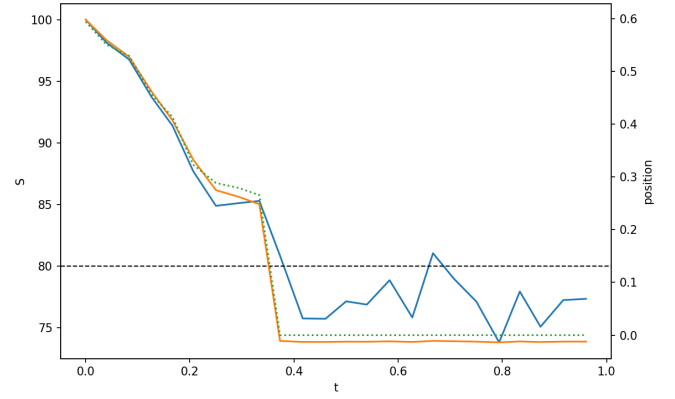
9. Cross-Experiment Discussion

9.1 What is actually validated?

The experiments have different scientific statuses. Experiment 1 is a strong numerical validation: the HJB solver reproduces a known closed form. Experiment 2 is an empirical stress test: it demonstrates parameter-estimation fragility rather than proving or disproving Merton optimality. Experiment 3B validates Bellman control in the frictionless European-call case and provides qualitative evidence of endogenous no-trade behavior with costs. Experiment 4 validates the frictionless barrier mechanism reasonably well but keeps the multidimensional cost branch exploratory.

9.2 From closed-form control to state-dependent hedging

The progression across experiments is structural. CRR homogeneity makes the Merton benchmark one-dimensional and analytically solvable. Hedging with proportional costs adds the current position as a state variable, while the barrier contract adds the survival indicator. The resulting increase in state dimension explains why the frictionless reductions are highly accurate whereas the cost branches become computationally demanding. At the same time, the experiments separate model optimality from empirical robustness: a control may be exact under known parameters yet fragile when those parameters must be estimated from data.

(a) Bellman and delta policies at $t = 0$.

(b) Example trajectory with knock-out behavior.

Figure 8: Experiment 4: barrier risk changes the feedback hedge, especially near the knock-out boundary.**Table 4:** Scientific status of the final experiments.

Experiment	Core question	Main result	Status
1. Merton HJB	Can FDM recover the analytic stochastic-control solution?	Value and feedback errors are very small; spatial convergence is clean.	Validated
2. AAPL Merton	Does the closed-form allocation remain robust with estimated inputs?	Long-only versions are stable but do not dominate benchmarks; unconstrained leverage is fragile.	Validated with limits
3A. Call heuristic	How do rebalancing and costs interact?	High frequency helps without costs; no-trade rules improve the cost/error compromise with costs.	Validated
3B. Call Bellman	Can Bellman discover the hedge?	At $\lambda = 0$, policy RMSE vs BS delta is 0.0031; HOLD emerges with costs.	Validated with limits
4. Barrier Bellman	Can control adapt to knock-out risk?	Bellman matches benchmark away from barrier and cuts exposure when knock-out becomes likely.	Exploratory/validated frictionless core

9.3 Physical and risk-neutral measures

The option-pricing benchmarks use the risk-neutral measure, replacing the physical drift by r , while the simulated hedging backtests and Bellman expectations use the physical drift $\mu = 8\%$. In the continuous frictionless Black–Scholes model, exact replication makes the delta independent of the physical drift. In discrete mean-square hedging, however, the objective is evaluated under the physical distribution, so the Bellman optimizer may differ slightly from the risk-neutral delta. This distinction explains why “matching the delta” is a validation target rather than an identity imposed in the code.

10. Limitations

The conclusions are subject to three main limitations. First, the market models are deliberately stylized: Merton and Black–Scholes assume lognormal dynamics with constant volatility, while the AAPL study uses a single asset and rolling historical estimators. In particular, drift estimation is noisy, and unconstrained CRRA allocations can leave the positive-wealth domain in which the utility model is meaningful.

Second, the transaction-cost Bellman problems are computationally limited by state dimension. The ex-

act quadratic-in-wealth reduction is highly effective at zero cost, but the implemented proportional-cost formulation requires explicit cash and position states. The reported BUY/HOLD/SELL regions are therefore meaningful structural diagnostics, whereas the corresponding multidimensional RMSE values should be interpreted at the stated grid resolution rather than as fully converged optima. This limitation is stronger for the barrier-cost pilot, whose state is (S, b, q, I) .

Third, the execution model is intentionally simple. Real trading costs also include bid–ask spreads, market impact, funding constraints, discrete lot sizes, latency, and stochastic liquidity. Brownian-bridge correction improves continuous-barrier event detection, but hedging decisions themselves remain discrete in time. Extending the framework to richer dynamics or execution models would require more scalable approximation methods.

11. Conclusion

This research project developed and tested a coherent stochastic-control pipeline for financial allocation and hedging. The Merton problem provided a controlled environment in which a finite-difference HJB method could be validated against an analytic value function and optimal

feedback. The agreement was excellent, establishing the numerical foundation of the project.

The empirical AAPL study then showed why a closed-form optimum is not automatically a robust trading rule: once the unknown drift and volatility are replaced by rolling estimates, leverage and parameter error become dominant concerns. This separates mathematical optimality under a model from out-of-sample performance in data.

The hedging experiments moved to genuine feedback control. For a European call without costs, discrete Bellman optimization independently recovered the Black–Scholes delta with policy RMSE about 3.1×10^{-3} . Proportional costs generated endogenous HOLD regions, as economic intuition predicts, although the multidimensional cost branch remains numerically imperfect. For a down-and-out call, the Bellman state was augmented by barrier survival and Brownian-bridge monitoring. The resulting policy matched standard deltas away from the barrier but reduced exposure sharply when knock-out became imminent.

The overarching conclusion is that stochastic control is not valuable because it reproduces formulas already known in simple models; those formulas are benchmarks. Its value appears when the decision must depend on the evolving state, transaction frictions, or path-dependent contract features. The project therefore progresses from verification to situations where the control policy itself becomes the object of interest.

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